



Date : Apr 2025	CIE – I	Max. Marks : 10 + 50
Semester : VI	UG	Duration : 30 + 90 Min
Course Title: Entrepreneurship & Intellectual Property Rights		Course Code : HS361TA

SCHEME & SOLUTIONS

Sl. No	Solutions with Scheme	M
Part – A		
1.	An innovative entrepreneur creates new ideas and inventions, whereas an imitative entrepreneur copies existing products or services with minimal modifications.	1
2.	Fabian entrepreneurs tend to be more cautious and resistant to change , which makes them less adaptive in fast-paced industries where innovation is crucial for survival.	1
3.	The main reason for failure is poor market research , leading to products or services that fail to meet customer needs or lack demand.	1
4.	Risk-taking is crucial because it allows entrepreneurs to pursue new opportunities and innovations, even when outcomes are uncertain, which can lead to high rewards.	1
5.	Tech-driven entrepreneurship , including e-commerce , software development , and digital platforms , has seen significant growth due to advancements in digital technologies.	1
6.	Entrepreneurship involves creating and running a new business, while intrapreneurship is about innovating within an existing organization, often with less risk and more resources.	1
7.	The primary purpose of IP laws is to protect creators' rights and incentivize innovation by granting exclusive rights to the creators or inventors.	1
8.	Public disclosure before filing typically destroys the novelty of the invention, making it non-patentable.	1
9.	A new pharmaceutical drug can be patented, as it meets the criteria for novelty and industrial applicability.	1
10.	Yes, patents are territorial , meaning the patent rights only apply in the country where the patent was granted.	1

**Part – B**

1 a.	Definition of Entrepreneurship (1 marks): Entrepreneurship is the process of designing, launching, and running a new business, often initially a small business, offering a product, process, or service for sale or hire. Focus on creating innovative solutions to address market needs and generating value. Scope of Entrepreneurship in Engineering (2 marks): • Involves identifying technological and market opportunities. • Key areas in engineering: Product development, process innovation, industrial automation, renewable energy, and engineering services. • Entrepreneurship fosters innovation, commercialization of new technologies, and economic growth through job creation. Key Areas of Entrepreneurial Impact (2 marks): • Startups in engineering sectors: Engineering-based companies such as those in robotics, biotechnology, IT solutions, etc. • Commercializing engineering inventions: Bringing ideas from concept to product, e.g., software applications, manufacturing equipment, and smart devices.	5
b.	Key Characteristics (2 ½ marks): • Risk-taking ability: Willing to take calculated risks to innovate. • Vision: Ability to envision future trends and make early strategic decisions. • Persistence: Overcoming obstacles in product development or market acceptance. • Adaptability: Responding to changes in market demands, technology, or competition. Traits Essential for Engineering Entrepreneurs (2 ½ marks): • Technical expertise: Deep understanding of engineering principles and innovation. • Leadership and management skills: Ability to lead teams and manage resources efficiently. • Problem-solving skills: Identifying and solving engineering challenges creatively.	5



2.	<i>The following answers are sample responses provided as a reference for evaluating the key concepts and points expected in the student's answer. These responses outline the core elements that should be addressed. Answers may vary, and alternative valid responses may be considered as long as they effectively address the core concepts of the question.</i> a. Key Factor in Failure (2 marks): poor marketing and unclear product positioning, could have addressed this by conducting thorough market research and identifying a clearer target audience early on. b. Pivot in Startups (2 marks): A pivot is a significant shift in a startup's business model or product offering in response to market feedback. FitTrack pivoted to focus on chronic health conditions to better address a specific market need and improve product-market fit. c. Effectiveness of Pivot (2 marks): The pivot was effective as it provided clearer value proposition and targeted a niche market. However, risks included potential market rejection or misalignment with customer needs. Alternatives could have included refining the original product instead of a complete pivot. d. Factors in Product Launch Success/Failure (2 marks): market research, product-market fit, clear value proposition, and effective marketing strategy. A mismatch in any of these can lead to failure. e. Securing Venture Capital (2 marks): Securing venture capital provides financial support and resources for growth. For FitTrack Innovations, it enabled them to expand their product range and scale operations, helping to recover and succeed after the pivot.	10
----	--	----



3	<i>The following answers are sample responses provided as a reference for evaluating the key concepts and points expected in the student's answer. These responses outline the core elements that should be addressed. Answers may vary, and alternative valid responses may be considered as long as they effectively address the core concepts of the question.</i> a. Key Principles of Entrepreneurship (2.5 marks): 'SunRay' Innovations demonstrates key entrepreneurial principles through: • Innovation: The development of a solar-powered water purification system is a creative solution to address water scarcity in rural areas. • Value Creation: By offering a sustainable and eco-friendly product, the company aims to improve public health and living conditions. • Problem-solving: Focuses on providing a solution to a critical issue—access to clean drinking water, especially in underserved regions. b. Entrepreneurial Opportunities (2.5 marks): Potential opportunities for 'SunRay' Innovations to overcome challenges include: • Strategic Partnerships: Collaborating with NGOs, government bodies, or large organizations that work on water sanitation projects could provide funding and market access. • Crowdfunding: Tapping into crowdfunding platforms could help secure funds from people who support eco-friendly, community-driven solutions. • Targeted Marketing: Focusing on niche markets such as rural communities, disaster-prone areas, and off-grid locations can create demand in under-served regions. c. Role of entrepreneurship in addressing social and environmental challenges (2.5 marks) • Sustainability: 'SunRay' uses solar-powered technology, reducing reliance on non-renewable energy. • Social Impact: It provides clean water to rural areas, improving health and quality of life. • Environmental Responsibility: The use of solar energy contributes to environmental sustainability, aligning with entrepreneurship's role in solving social and environmental issues. d. Managing Risks of Expansion (2.5 marks): To effectively manage risks during expansion, 'SunRay' Innovations can: • Conduct Market Research: Understanding market needs and customer preferences in new areas will reduce the risk of product misalignment. • Scalable Production: Develop scalable and cost-effective manufacturing processes to ensure that demand can be met without significant cost overruns. • Diversification: Exploring different distribution channels (such as direct-to-consumer, partnerships, and government contracts) can reduce reliance on one source of revenue and mitigate risks.	10
---	---	----



4.a	Definition of Intellectual Property (IP) (1 mark): Intellectual Property refers to creations of the mind, such as inventions, literary and artistic works, designs, symbols, names, and images used in commerce. It grants the creator exclusive rights over the use of their creations. Importance of IP in the Modern Economy (2 marks): any 2 • Encourages Innovation: IP protections incentivize creators and innovators by providing exclusive rights to their inventions and creations. • Economic Growth: IP drives investment in research and development, fostering business growth and contributing to national economic prosperity. • Competitive Advantage: Businesses with strong IP portfolios have market differentiation, giving them an edge over competitors and enhancing brand value. Types of Intellectual Property and Their Significance for Businesses (2 marks): • Patents: Protect inventions, giving businesses exclusive rights to manufacture or use the invention. • Trademarks: Protect distinctive signs, logos, or names used to distinguish goods and services. Helps build brand identity. • Copyrights: Protect artistic and literary works, ensuring creators retain control over their creations. • Trade Secrets: Protect confidential business information, such as formulas or processes, crucial for maintaining a competitive edge.	5
4.b	Overview of the Patent Application Procedure (1 mark): write-up / flowchart The patent application procedure is a legal process through which an inventor seeks exclusive rights to their invention. A granted patent allows the inventor to exclude others from making, using, or selling the invention for a specified period (usually 20 years). Key Steps in Securing a Patent (3 marks): • Step 1: Patent Search: Conduct a thorough search to ensure the invention is novel and does not infringe on existing patents. • Step 2: Prepare and File the Patent Application: Prepare a detailed patent application, including claims describing the invention, a description of how it works, and drawings if applicable. Submit the application to the relevant patent office. • Step 3: Examination: The patent office examines the application to determine if the invention is novel, non-obvious, and industrially applicable. The examiner may issue office actions requesting clarifications or amendments. • Step 4: Grant or Refusal: If the application meets all criteria, a patent is granted. If not, the application may be refused or the inventor can amend it and reapply. Timelines for the Patent Process (1 mark): • Filing to Examination: Typically, the examination begins within 6 to 12 months of filing, depending on the patent office. • Examination to Grant: The entire process, from filing to grant, can take anywhere from 1 to 5 years, depending on the complexity of the patent and the workload of the patent office. • Possible Delays: Delays may occur due to backlogs, additional clarifications, or complications in the patent examination.	5



5	<i>The following answers are sample responses provided as a reference for evaluating the key concepts and points expected in the student's answer. These responses outline the core elements that should be addressed. Answers may vary, and alternative valid responses may be considered as long as they effectively address the core concepts of the question.</i> a. Importance of Conducting a Patent Search (2.5 marks): • Avoids Infringement: A patent search helps InnovTech identify if similar inventions already exist, preventing infringement on existing patents. • Ensures Novelty: It ensures the uniqueness of the invention, helping InnovTech confirm that their technology is novel, a key requirement for patentability. • Strategic Decision-Making: Identifying similar patents early on allows InnovTech to adjust their technology, claims, or business strategy to differentiate their product. b. Steps to Protect Invention and Ensure Patentability (2.5 marks): • Document the Invention: InnovTech should thoroughly document the development process to prove originality and priority over the invention. • File for Patent Early: InnovTech should continue to pursue patent filing in their home country (India) and ensure they meet all patentability criteria—novelty, non-obviousness, and industrial applicability. • Consider Patent Claims: InnovTech should draft strong patent claims that highlight the unique aspects of their device, focusing on the efficiency and scalability of their technology. • Conduct Further Research: InnovTech should analyze the competing patents to understand potential overlaps and strengthen their claims or modify their technology. c. Steps for Global Patent Protection (2.5 marks): • Patent Cooperation Treaty (PCT): InnovTech can file a PCT application to seek international protection for their invention in multiple countries. • Regional Patent Systems: They can also explore regional patent systems, such as the European Patent Office (EPO), to file for protection in multiple countries in one go. • National Patent Filings: Alternatively, InnovTech can directly file patent applications in individual countries, ensuring compliance with each jurisdiction's patent laws. d. Options for Transferring or Commercializing the Patent (2.5 marks): • Licensing: InnovTech can license their patent to larger corporations, granting them the right to use the technology in exchange for royalty payments. • Patent Assignment: InnovTech could transfer the patent rights entirely to another entity, thereby receiving a lump sum payment for the sale. • Joint Ventures or Partnerships: Collaborating with other companies for mutual commercialization and development of the technology. • Legal Remedies for Infringement: If InnovTech's patent rights are infringed upon, they can seek injunctive relief (stop the infringement), claim damages (compensation for losses), or file a patent infringement lawsuit in the relevant jurisdiction to protect their intellectual property.	10
---	--	----